

Defining Fractal Dimension

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Introduction

In this paper we examine a notion of the size of a set called *fractal dimension* that enables one to more finely distinguish the density of a given set than the usual dimensional classification of $0, 1, 2, 3, \dots$ ¹ Such a notion is motivated by sets that, while constructed in one of the standard dimensions $0, 1, 2, 3, \dots$ and robust in their structure, nevertheless have zero “volume” in the ambient space containing them. Such sets beg for a more nuanced “size” metric.

The sets that we are interested in are non-physical sets that have the property that they retain structure no matter how much they are magnified. We call this class of sets *fractal sets*, or *fractals*.

A commonplace example that resembles a fractal is the coastline of a beach. If one were to take a map of the coastline and blow it up again and again, one would find that it retains structure in the sense that the expanded maps would not “straighten out”. Because this is a physical example, the complicated structure would fade after a finite number of expansions. A true fractal, say representing a coastline, would retain complicated structure no matter how many times a piece of it was magnified. Because of this, the length of any section of this idealized coastline is infinite. Another “natural” fractal is the surface of an idealized sponge. While defined in a three dimensional world, one can construct such a sponge so that it retains complicated structure no matter how many times it is magnified. In addition, such a constructed sponge could have an infinite surface area and also have 0 volume.

It would be convenient if there were a way to measure the density of an idealized coastline to indicate that it is, in some sense, larger than a one dimensional object; or a way to show that the idealized surface of a sponge is larger than two dimensions. This would allow us to compare the “size” of such sets in a more precise way.

This is what the fractal dimension of a set tries to address. In the case of the coastline, it will give us a dimension between one and two – where the coastline “naturally” lives. The same holds for the surface of the aforementioned sponge: there will be a dimension between two and three where it naturally lives. In each case, the fractal dimension is not an actual dimension, it is a way to quantify the density of such complex sets. Yet, it should also “agree” with the usual dimensional size of an object if it is not a fractal. That is, the fractal dimension of a line segment should be 1; the fractal dimension of a filled triangle (its interior together with its boundary) sitting in three dimensions should be 2, etc.

When studying any new phenomenon, it is natural to first examine the simplest examples. The simplest such fractals are ones which have a *self-similarity* property; namely, they have the property that if one magnifies a piece of the fractal by a certain amount, the original fractal is recovered. This regular structure makes them easier to study mathematically than a general fractal. However, before we start with more specific examples of fractals, we need to understand what one might mean by fractal dimension.

¹ The 0-dimensional volume of a finite set is its cardinality; for an infinite set it is $+\infty$.

Volume of an n-Dimensional Sphere

The n-dimensional volume of any sphere is easy to compute if one knows its surface area. Since the surface area of an n-dimensional sphere with radius, r , is r^{n-1} times the surface of the unit sphere, it suffices to know the surface area of the unit sphere. In what follows we will refer to an n-dimensional sphere of radius, r , as $B_n(r)$, and B_n as the specific n-dimensional sphere of radius 1. We will use the notation, $S_n(r)$ and $V_n(r)$ for the surface area and volume of $B_n(r)$ respectively.

The relationship between $V_n(r)$ and $S_n(1)$ follows from the following formula:

$$V_n(r) = \int_{\partial B_n} \int_0^r \tau^{n-1} d\tau dS = \left(\int_{\partial B_n} dS \right) \left(\int_0^r \tau^{n-1} d\tau \right)$$

The iterated integrals on the right hand side of the above equation can be easily computed, yielding: $V_n(r) = S_n(1) \frac{r^n}{n}$.

What remains now is to find a formula for $S_n(r)$. This can be done by expressing an easily computable n-dimensional integral in R^n as n iterated integrals. And more importantly, this integral may be represented as a *different* spherically symmetric integral involving the surface area of B_n .

$$\int_{-\infty}^{\infty} e^{-(x_1^2+x_2^2+\dots+x_n^2)} dx_1 dx_2 \dots dx_n = \int_{\partial B_n} \int_0^{\infty} e^{-r^2} r^{n-1} dr dS$$

The left side of the above can be written as an iterated integral of identical one dimensional integrals. The right side of the equation is the same integral but written in “spherical coordinates” which is a convenient choice as the integrand is spherically symmetric. The left side simplifies to $\int_{-\infty}^{\infty} e^{-x^2} dx$ raised to the n^{th} power; which is $\pi^{n/2}$. Substituting $t = r^2$ in the inner integral on the right hand side we have:

$$\pi^{n/2} = \frac{1}{2} \int_{\partial B_n} \int_0^{\infty} e^{-t} t^{\frac{n}{2}-1} dt dS$$

which may be written:²

$$2\pi^{n/2} = S_n(1)\Gamma(n/2)$$

Or,

$$S_n(1) = \frac{2\pi^{n/2}}{\Gamma(n/2)}$$

Consequently,³

$$S_n(r) = \frac{2\pi^{n/2}}{\Gamma(n/2)} r^{n-1}$$

And so the volume of an n-dimensional sphere, $B_n(r)$, of radius r is:

$$V_n(r) = \int_{B_n(r)} d\mu = \int_{\partial B_n} \int_0^r \tau^{n-1} d\tau dS = \left(\frac{r^n}{n} \right) S_n(1)$$

Which is

$$V_n(r) = \frac{2\pi^{n/2}}{n \Gamma(n/2)} r^n$$

² The function $\Gamma(z)$ is called the Gamma function. It is defined for all real (and complex) arguments *except* the non-positive integers $0, -1, -2, \dots$, and satisfies $\Gamma(z+1) = z\Gamma(z)$ wherever defined. On the positive integers it agrees with the factorial: $\Gamma(n+1) = n!$ for $n \in \mathbb{Z}^+$. In this sense Γ extends the factorial from the integers to the reals.

³ The surface area of an n-dimensional sphere scales by the radius to the $(n-1)^{\text{th}}$ power.

Fractal Dimension and Associated Volume of a Set

We derived the formula for the n -dimensional volume of a sphere, and the formula may take non-integral values. Armed with this formula, we make the following definition.

Definition. For any $\alpha \in R^+$, the α -dimensional volume, V_α , of an n -dimensional sphere of radius r is defined to be:⁴

$$V_\alpha(r) = \frac{2\pi^{\alpha/2}}{\alpha \Gamma(\alpha/2)} r^\alpha = \frac{\pi^{\alpha/2}}{\Gamma(\alpha/2+1)} r^\alpha$$

The second form uses the identity $\alpha \Gamma(\alpha/2) = 2 \Gamma(\alpha/2 + 1)$, and shows that – despite appearances – the formula has no singularity as $\alpha \rightarrow 0^+$: the apparent poles of $1/\alpha$ and $\Gamma(\alpha/2)$ cancel, and the limiting value is $V_0(r) = 1$. This is consistent with the counting-measure convention that the 0-dimensional volume of a single point is 1. We nevertheless restrict to $\alpha > 0$ in what follows.

This is a mathematical construct, not a physical one; there is no claim here of a physical fractional dimension. Also, note that the value is independent of the ambient space in which the ball might live, say R^n . That is, α may be taken to be larger or smaller than n .

Given that we have defined the fractional dimensional volume of a sphere in “dimension” α we wish to extend this to a definition of the fractional dimensional volume of *any* set.

Definition. The α -dimensional ($\alpha \in R^+$) covering volume of a set $S \subseteq R^n$, $\mathcal{V}_{\alpha,\epsilon}(S)$, with parameter $\epsilon \in R^+$, is defined as⁵

$$\mathcal{V}_{\alpha,\epsilon}(S) = \inf_{\{O_i\}_{i=1}^N} \left\{ \sum_{i=1}^N V_\alpha(\text{Diam}(O_i)/2) : S \subseteq \bigcup_{i=1}^N O_i \wedge \left(\sup_{i \in [1,N]} \text{Diam}(O_i) \right) \leq \epsilon \right\}$$

Roughly speaking, for any given finite covering, $\{O_i\}_{i=1}^N$, and value ϵ , we estimate the α -dimensional volume of the set, S , by charging each covering set O_i the α -dimensional volume of a sphere of radius $\text{Diam}(O_i)/2$. That is, each summand may be written

$$V_\alpha(\text{Diam}(O_i)/2) = c_\alpha \text{Diam}(O_i)^\alpha, \quad c_\alpha = \frac{2^{1-\alpha} \pi^{\alpha/2}}{\alpha \Gamma(\alpha/2)}.$$

The factor c_α is a normalization, chosen so that for integer α the covering volume of ordinary solids agrees with their usual volume. It is *not* literally the volume of the smallest sphere containing O_i : a set of diameter d need not fit inside a sphere of radius $d/2$ (by Jung’s theorem the smallest enclosing sphere in R^n can have radius as large as $d\sqrt{n/(2n+2)}$). The α covering volume of a set, S , is the infimum over all such covers. The only restrictions on the covering sets are that they be finite and that each have diameter no larger than ϵ .

Remark. The standard Hausdorff-measure definition differs from the one above in one respect: it allows *countably infinite* covers, and in general that is the definition one should use. For the compact sets treated in this note the two definitions give the same answers, but they can differ on non-closed or unbounded sets. For example, any finite cover of the set of rationals in $[0, 1]$ by sets of diameter at most ϵ has closures that cover all of $[0, 1]$, so under finite covers the rationals in $[0, 1]$ receive dimension 1; under countable covers – each rational covered by its own arbitrarily small interval – they receive dimension 0. Likewise, an unbounded set admits no finite cover at all. We keep finite covers here for concreteness, since our examples are compact.

⁴ The set R^+ is defined by: $R^+ = \{x \mid x \in R \wedge x > 0\}$.

⁵ The diameter of a set is defined by: $\text{Diam}(S) = \sup\{|x - y| : x, y \in S\}$.

The intent of this definition is to give a practicable estimate of the eventual α volume of a set, refined by forcing the covers to become finer as ϵ decreases.

The covering volume is monotonically non-increasing in ϵ :

$$\mathcal{V}_{\alpha, \epsilon_1}(S) \geq \mathcal{V}_{\alpha, \epsilon_2}(S) \quad \text{whenever } \epsilon_1 \leq \epsilon_2$$

This follows as the set of coverings used to compute $\mathcal{V}_{\alpha, \epsilon_1}(S)$ is a *subset* of those used to compute $\mathcal{V}_{\alpha, \epsilon_2}(S)$ (a tighter ϵ admits fewer covers). Therefore, the infimum over the smaller family cannot be smaller than the infimum over the larger family.

Using the covering volume, we may define the fractional dimensional volume by taking a limit of increasingly refined covers.

Definition. The α -dimensional ($\alpha \in R^+$) volume of a set $S \subseteq R^n$, $\mathcal{V}_\alpha(S)$, is defined as⁶

$$\mathcal{V}_\alpha(S) = \lim_{\epsilon \rightarrow 0^+} \mathcal{V}_{\alpha, \epsilon}(S)$$

While these statements define the α volume for any given set, S , they do not tell us the “natural” fractal volume of S . The following definition does precisely that.

Definition. The **fractal dimension** of a set $S \subseteq R^m$ is defined as $\alpha^* = \inf\{\alpha : \mathcal{V}_\alpha(S) = 0\}$; the associated **fractal volume** of S is defined to be $\mathcal{V}_{\alpha^*}(S)$.

That α^* is a genuine threshold – with the α volume zero above it and infinite below it – is the content of the following proposition.

Proposition. Let $S \subseteq R^n$ and $0 < \alpha < \beta$. If $\mathcal{V}_\alpha(S) < \infty$, then $\mathcal{V}_\beta(S) = 0$. Equivalently, if $\mathcal{V}_\beta(S) > 0$, then $\mathcal{V}_\alpha(S) = +\infty$.

Proof. For any admissible cover, each set O_i has $\text{Diam}(O_i) \leq \epsilon$, so

$$\sum_{i=1}^N c_\beta \text{Diam}(O_i)^\beta \leq \frac{c_\beta}{c_\alpha} \epsilon^{\beta-\alpha} \sum_{i=1}^N c_\alpha \text{Diam}(O_i)^\alpha$$

Taking the infimum over covers gives $\mathcal{V}_{\beta, \epsilon}(S) \leq (c_\beta/c_\alpha) \epsilon^{\beta-\alpha} \mathcal{V}_{\alpha, \epsilon}(S) \leq (c_\beta/c_\alpha) \epsilon^{\beta-\alpha} \mathcal{V}_\alpha(S)$, where the last step uses the monotonicity of $\mathcal{V}_{\alpha, \epsilon}$ in ϵ . Letting $\epsilon \rightarrow 0^+$ yields $\mathcal{V}_\beta(S) = 0$.

Consequently, for $\alpha > \alpha^*$ the α volume of S is zero, and for $\alpha < \alpha^*$ it is infinite. At $\alpha = \alpha^*$ itself the volume may be zero, a positive finite number, or infinite; when it is positive and finite – as in the examples below – it serves as a natural “size” of S in its natural dimension.

The quantity α^* is precisely the *Hausdorff dimension* of S (subject to the finite-cover caveat of the previous remark); see Falconer [1] for the standard treatment. The term *fractal* was coined by Mandelbrot [2].

Note that the fractal dimension of a set, S , does not depend on the ambient space S sits in.

Computing the Fractal Dimension of a Simple Set

⁶ This definition is well defined, as the limit of a monotonic sequence exists, provided we acknowledge that it may be infinite.

The fractal dimension of a set S should produce the usual answer when S is a non-fractal set. We show this for the case of a straight line.

Example: A line segment in two dimensions.

Let $\mathbf{x}, \mathbf{y} \in R^2$ and consider the line segment

$$L = \{t\mathbf{x} + (1-t)\mathbf{y} : t \in [0, 1]\}.$$

We wish to find the fractal dimension and fractal volume of L . While this line segment sits in the two dimensional plane, we wish to find its “natural” dimension and associated volume.

For a given n , consider the sets $\{O_{n,i}\}_{i=0}^{n-1}$, with $O_{n,i} = [\frac{i}{n}\mathbf{x} + (1 - \frac{i}{n})\mathbf{y}, \frac{i+1}{n}\mathbf{x} + (1 - \frac{i+1}{n})\mathbf{y}]$ denoting the line segment in R^2 between the two endpoints in brackets. The collection of line segments, $O_{n,i}$, act as a cover for L ; that is, $L = \bigcup_{i=0}^{n-1} O_{n,i}$. For a given α the covering volume of L with $\epsilon = \frac{1}{n}\|\mathbf{x} - \mathbf{y}\|$ is bounded above by

$$\sum_{i=0}^{n-1} V_\alpha(\text{Diam}(O_{n,i})/2) = \sum_{i=0}^{n-1} V_\alpha(\|\mathbf{x} - \mathbf{y}\|/(2n)) = \sum_{i=0}^{n-1} \frac{2\pi^{\alpha/2}}{\alpha\Gamma(\alpha/2)} \frac{\|\mathbf{x} - \mathbf{y}\|^\alpha}{(2n)^\alpha} = \frac{2\pi^{\alpha/2}\|\mathbf{x} - \mathbf{y}\|^\alpha}{2^\alpha \alpha\Gamma(\alpha/2)} \frac{n}{n^\alpha}$$

As n increases, the covering becomes more refined and the upper bound is dominated by the term $\frac{n}{n^\alpha} = n^{1-\alpha}$. For $\alpha > 1$ this tends to zero as $n \rightarrow \infty$, so $\mathcal{V}_\alpha(L) = 0$. For $\alpha = 1$ the bound equals $c_1\|\mathbf{x} - \mathbf{y}\|$ with $c_1 = \pi^{1/2}/\Gamma(1/2) = 1$, so $\mathcal{V}_1(L) \leq \|\mathbf{x} - \mathbf{y}\|$.

These covers give only *upper* bounds; to pin down the dimension we need a matching lower bound valid for *every* cover. Let $\{O_i\}_{i=1}^N$ be any cover of L with $d_i = \text{Diam}(O_i) \leq \epsilon$. Orthogonal projection onto the line through $\mathbf{x}$ and $\mathbf{y}$ does not increase diameters, and the projections of the O_i cover the segment L , so

$$\sum_{i=1}^N d_i \geq \|\mathbf{x} - \mathbf{y}\|.$$

For $\alpha \leq 1$ we have $d_i^\alpha = d_i^{\alpha-1} d_i \geq \epsilon^{\alpha-1} d_i$ (as $d_i \leq \epsilon$ and $\alpha - 1 \leq 0$), whence

$$\sum_{i=1}^N V_\alpha(d_i/2) = c_\alpha \sum_{i=1}^N d_i^\alpha \geq c_\alpha \epsilon^{\alpha-1} \|\mathbf{x} - \mathbf{y}\|.$$

For $\alpha < 1$ the right side tends to $+\infty$ as $\epsilon \rightarrow 0^+$, so $\mathcal{V}_\alpha(L) = +\infty$. For $\alpha = 1$ it gives $\mathcal{V}_{1,\epsilon}(L) \geq \|\mathbf{x} - \mathbf{y}\|$, which combined with the upper bound above yields

$$\mathcal{V}_1(L) = \|\mathbf{x} - \mathbf{y}\|.$$

Therefore, the fractal dimension of the line segment is 1 and its fractal volume is the length of the line segment, $\|\mathbf{x} - \mathbf{y}\|$. This, of course, coincides with what we would previously regard as the dimension and “volume” of a line segment.

Note that the answer would stay the same if the line segment was embedded in R^n (for any $n \in \mathbb{Z}^+$) rather than R^2 .

This result is true for any “ordinary” set: its fractal dimension is its “ordinary” dimension. This means that we can think of fractal dimension as a *generalization* of dimension. It is instructive to repeat the above analysis and compute the fractal dimension and fractal volume of a simple fractal.

The Cantor Set: A Simple Fractal Set

Perhaps the simplest self-similar fractal is the Cantor set, which is formed by doing the following: Take the unit interval, $[0,1]$, and remove the middle third, $(1/3, 2/3)$. This leaves the intervals, $[0,1/3]$ and $[2/3,1]$. Now remove the middle third of each of these two intervals, this gives four intervals. If we repeat this process indefinitely with the remaining intervals, the resulting set is called the Cantor set, C . See the appendix for a graphical depiction of the construction process. The Cantor set has the property that $3 \cdot ([0, 1/3] \cap C) = C$; that is, it is a self-similar fractal (where $\lambda \cdot S \equiv \{\lambda x : x \in S\}$).

The one dimensional length of the Cantor set is easy to compute, it must be one (the length of the interval $[0,1]$) less the sum of all the lengths of the intervals removed. This is computed as $1 - \sum_{n=1}^{\infty} (2^{n-1}/3^n) = 1 - \frac{1}{2} \sum_{n=1}^{\infty} (2/3)^n = 0$. Since the Cantor set has zero length, perhaps its zero dimensional “length” is a more natural description of its size. But its zero dimensional length is equal to the number of points in the Cantor set; and, it turns out that this is infinite. What’s more startling is that, in a sense that can be made precise, there are as many points in the Cantor set as there are in the interval $[0,1]$. So, although the Cantor set has no length, it is comparable in complexity to $[0,1]$. This suggests that it might have a natural dimension between zero and one.

The Fractal Dimension of the Cantor Set and its Volume

We compute the fractal dimension of the Cantor set. To do so, consider a refinement of coverings of the Cantor set, C_n , where C_n is the union of the set of 2^n intervals that remain after the n^{th} stage of the Cantor construction process. Each of these intervals has length $(1/3)^n$. Clearly, $C \subseteq C_n$, and $C = \bigcap_{n=0}^{\infty} C_n$ (with $C_0 = [0,1]$).

We label $I_{n,i}$ as the i^{th} interval in the set C_n . Each of these intervals has a diameter, $\epsilon_n = (\frac{1}{3})^n$. The collection, $\bigcup_{i=1}^{2^n} I_{n,i}$, provides an approximation to $\mathcal{V}_{\alpha, \epsilon_n}(C)$:

$$\mathcal{V}_{\alpha, \epsilon_n}(C) \leq \sum_{i=1}^{2^n} V_{\alpha}(\text{Diam}(I_{n,i})/2)$$

As $\text{Diam}(I_{n,i}) = (\frac{1}{3})^n$, this becomes

$$\mathcal{V}_{\alpha, \epsilon_n}(C) \leq \sum_{i=1}^{2^n} \frac{2\pi^{\alpha/2}}{\alpha\Gamma(\alpha/2)2^{\alpha}3^{n\alpha}} = \frac{2^{1-\alpha}\pi^{\alpha/2}}{\alpha\Gamma(\alpha/2)} \sum_{i=1}^{2^n} \frac{1}{3^{n\alpha}}$$

Or,

$$\mathcal{V}_{\alpha, \epsilon_n}(C) \leq \frac{2^{1-\alpha}\pi^{\alpha/2}}{\alpha\Gamma(\alpha/2)} \left(\frac{2}{3^{\alpha}}\right)^n$$

If $\frac{2}{3^{\alpha}} < 1$ then, in the limit as $n \rightarrow \infty$, this upper bound will be zero. This is the case exactly when $\alpha > \ln 2 / \ln 3$; in which case

$$\mathcal{V}_{\alpha}(C) = \lim_{n \rightarrow \infty} \mathcal{V}_{\alpha, \epsilon_n}(C) \leq \lim_{n \rightarrow \infty} \frac{2^{1-\alpha}\pi^{\alpha/2}}{\alpha\Gamma(\alpha/2)} \left(\frac{2}{3^{\alpha}}\right)^n = 0$$

For $\alpha < \ln 2 / \ln 3$ the same expression diverges ($\frac{2}{3^{\alpha}} > 1$), so these covers no longer force the volume to zero. In fact $\mathcal{V}_{\alpha}(C) = +\infty$ in that range: by the Proposition, this follows from the positivity of the volume at the threshold itself, $\mathcal{V}_{\alpha^*}(C) > 0$, discussed below.

The threshold value of α – at which $\frac{2}{3^\alpha} = 1$ – occurs when $\alpha^* = \ln 2 / \ln 3$. Therefore, the fractal dimension of the Cantor set is $\alpha^* = \ln 2 / \ln 3 \approx 0.63093$. Its fractal volume in this dimension is bounded above by $\mathcal{V}_{\alpha^*}(C) \leq \frac{2^{1-\alpha^*} \pi^{\alpha^*/2}}{\alpha^* \Gamma(\alpha^*/2)} = c_{\alpha^*} \approx 1.03505$. A matching lower bound is genuinely harder: one must show that *no* sequence of covers can do better than the natural covers used above. This is a classical fact: in the standard normalization – where a cover set of diameter d is charged the amount d^α rather than $c_\alpha d^\alpha$ – the α^* -dimensional measure of the Cantor set is exactly 1; see Falconer [1], where the lower bound is obtained by a mass distribution argument that exploits the self-similarity of C . In our normalization this gives

$$\mathcal{V}_{\alpha^*}(C) = c_{\alpha^*} \approx 1.03505.$$

Conclusion

We have shown that there are sets with sufficient complexity that the usual categorization of dimension as a measure of size is lacking. We have further shown that the notion of fractal dimension, as an extension to the usual notion of dimension, provides a more discriminating view of the size of a set.

While physical examples like a coastline or the surface of a sponge do not really have complexity on an infinite scale, we idealize them as such with fractals. Just as “dimensionless” points are an idealization useful to describe space, fractals are also a useful idealization. As with all such mathematical abstractions, they allow us to view and compute what they model in a simpler way.

References

- [1] K. J. Falconer, *The Geometry of Fractal Sets*, Cambridge University Press, 1985.
- [2] B. B. Mandelbrot, *The Fractal Geometry of Nature*, W. H. Freeman, 1982.

Appendix A: Examples of Self-Similar Fractals

We briefly describe two *self-similar* fractal sets. The first set lives in 1 dimension, but its fractal dimension is between 0 and 1. The set is called the Cantor set, named after the mathematician Georg Cantor. This set is constructed by defining intermediate sets $C_0, C_1, \dots$ and setting the Cantor set, C , to be the intersection of these sets:

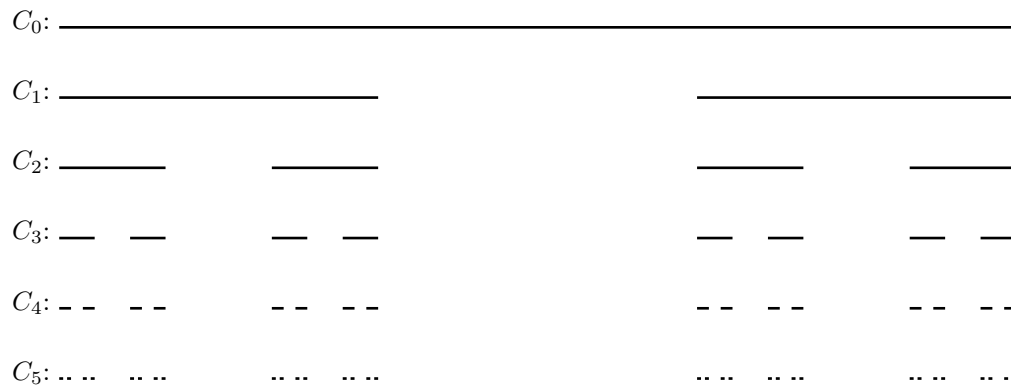
$$C = \bigcap_{n=0}^{\infty} C_n$$

The intermediate sets have the property that $\forall n, C_{n+1} \subseteq C_n$. These sets are defined by:

$$C_0 = [0, 1]$$

C_{n+1} = For each contiguous segment of the set C_n , remove the open interval that is its middle third.

As the sets, C_n , are nested, each successive set is an improved approximation to the Cantor set. The first 6 intermediate sets are graphed below:



The second self-similar set lives in 2 dimensions while its fractal dimension is between 1 and 2. It is the *Sierpiński triangle* (a different object from the 3-dimensional Menger sponge mentioned in the introduction, although both are sometimes loosely called “sponges”); it is “best” described in a recursive way as a scaled version of the lower triangular portion of itself, and its fractal dimension is $\ln 3 / \ln 2 \approx 1.585$.

Similar to the graphical display of the Cantor set construction, below we show a graph of 8 “levels” of its construction without going into construction details.

